

The Economics of Bicycles for the Mind

Agrawal, Gans & Goldfarb (2025)

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<https://github.com/carlosgomezp28/ai-02-agrawal>

Repository 2 — Artificial Intelligence and Economic Modeling

The Paper: Cognitive Tools and Human Capabilities

Question. How do cognitive tools change productivity and the value of different human capabilities?

Implementation skill s

- ▶ Converts effort into successful implementation
- ▶ Tools substitute for it

Agent's problem

Opportunity judgment ρ

- ▶ Recognizes further improvements
- ▶ Complements tools

Payoff judgment η

- ▶ Turns success into value
- ▶ Complementarity is conditional

$$M(e; \theta) = p(se; \theta)\eta\Delta - c(e; \theta), \quad e^*(\theta) \in \arg \max_{e \geq 0} M(e; \theta)$$

$$p_x(se; \theta)s\eta\Delta = c_e(e; \theta).$$

AI improves implementation technology, but the worker optimally changes effort in response.

Main Results: Propositions 1–3

Proposition 1

$$e^*(1) < e^*(0), \\ V_0(1) > V_0(0).$$

Less implementation effort,
greater total value.

Proposition 2

- ▶ Opportunity judgment:
complement
- ▶ Implementation skill:
substitute if $p_{s\theta} < 0$
- ▶ Payoff judgment: conditional
complement

$$\frac{dD}{d\eta} = \Gamma \Delta [p(se^*(1); 1) \\ - p(se^*(0); 0)].$$

Proposition 3

$$e(\theta) = \frac{\eta^2 s}{4} - \frac{\theta}{s},$$

Interiority: $\eta^2 s^2 > 4\theta$.
$$V(\theta) = G \left[\frac{\eta^2 s}{4} + \frac{\theta}{s} \right].$$

Condition (30): inequality initially falls

$$\frac{E[G^2]}{(E[G])^2} < \bar{s} E[1/s].$$

What I Did

1. Reconstructed the effort FOC and Proposition 2 envelope-theorem argument.
2. Derived Proposition 3's variance from first principles:

$$\text{Var}(V(\theta)) = A + a_0\theta + \theta^2 \text{Var}(G/s), \quad \frac{d \text{Var}(V(\theta))}{d\theta} = a_0 + 2\theta \text{Var}(G/s).$$

3. Separated total-value inequality from inequality in individual AI benefits:

$$B_i(\theta) = V_i(\theta) - V_i(0) = \theta \frac{G_i}{s_i}, \quad \text{Var}(B(\theta)) = \theta^2 \text{Var}(G/s).$$

4. Audited the assumptions behind the claimed U-shape.

Takeaway

Unequal AI benefits can increase even while total outcome inequality initially falls.

Where I Did Not Believe AI

Initial LLM claim: “Variance is globally U-shaped in θ .”

My check

$$e(\theta) = \frac{\eta^2 s}{4} - \frac{\theta}{s} \quad \text{requires} \quad \eta^2 s^2 > 4\theta.$$

Therefore the interior formula cannot automatically be extended to all $\theta \geq 0$.

$$\frac{d^2 \text{Var}(V(\theta))}{d\theta^2} = 2 \text{Var}(G/s) \geq 0.$$

- ▶ Convex on the common interior domain.
- ▶ Condition (30) makes the initial slope negative.
- ▶ A U-shape additionally requires θ^* inside that domain.

Verdict: referee concern, not a confirmed paper error

Eq. (32)’s positive slope at $\theta = 1$ is not implied by condition (30) alone.

The suspected coefficient-of-variation typo was retracted because PDF extraction may have omitted a square root.



Handwritten
envelope-theorem check